



3 Complex Analysis 2024

Tutorial 2

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Limits

Problem 6 If the limit exists, evaluate it. If the limit does not exist, prove this.

$$(a) \lim_{z \rightarrow 1} \frac{\bar{z} - 1}{z^2 - 1}, \quad (b) \lim_{z \rightarrow i} \frac{z - i}{z^2 + 1}.$$

Solution.

- (a) The limit $\lim_{z \rightarrow 1} \frac{\bar{z} - 1}{z^2 - 1}$ does not exist. Indeed, we take $z = 1 + h$ with $h \in \mathbb{C}$, $h \rightarrow 0$ to rewrite the limit as

$$\lim_{z \rightarrow 1} \frac{\bar{z} - 1}{z^2 - 1} = \lim_{h \rightarrow 0} \frac{\overline{(1 + h)} - 1}{(1 + h)^2 - 1} = \lim_{h \rightarrow 0} \frac{1 + \bar{h} - 1}{1 + 2h + h^2 - 1} = \lim_{h \rightarrow 0} \frac{\bar{h}}{2h + h^2}.$$

Now take $h \in \mathbb{R}$, $h \rightarrow 0$, then

$$\lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{\bar{h}}{2h + h^2} = \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{h}{2h + h^2} = \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{1}{2 + h} = \frac{1}{2}.$$

On the other hand, if $h = ik$ with $k \in \mathbb{R}$, $k \rightarrow 0$, then

$$\lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{\overline{ik}}{2ik + (ik)^2} = \lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{-ik}{2ik - k^2} = \lim_{\substack{k \rightarrow 0 \\ k \in \mathbb{R}}} \frac{-i}{2i - k} = -\frac{1}{2}.$$

It follows that the limit does not exist.

- (b) We have

$$\lim_{z \rightarrow i} \frac{z - i}{z^2 + 1} = \lim_{z \rightarrow i} \frac{z - i}{z^2 - i^2} = \lim_{z \rightarrow i} \frac{z - i}{(z - i)(z + i)} = \lim_{z \rightarrow i} \frac{1}{z + i} = \frac{1}{2i} = -\frac{i}{2}.$$

Differentiability, holomorphic functions

Problem 7 Show from the definition of the derivative that the polynomial

$$f(z) = z^2 + 4iz - 4$$

is differentiable at $z = i$, and calculate the derivative $f'(i)$.

Solution.

We consider the limit

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{f(i+h) - f(i)}{h} &= \lim_{h \rightarrow 0} \frac{(i+h)^2 + 4i(i+h) - 4 - [i^2 + 4i \cdot i - 4]}{h} \\ &= \lim_{h \rightarrow 0} \frac{i^2 + 2ih + h^2 + 4i^2 + 4ih - 4 - i^2 - 4i^2 + 4}{h} \\ &= \lim_{h \rightarrow 0} (2i + 4i + h) = 6i.\end{aligned}$$

The limit exists; this means that f is differentiable at $z = i$ and $f'(i) = 6i$.

Problem 8 We know that the function $f(z) = z^2 + 4iz - 4$ is differentiable at any point because it is a polynomial. Show that it satisfies the Cauchy-Riemann equations at any point.

Solution.

Let $z = x + iy$, then

$$f(z) = (x + iy)^2 + 4i(x + iy) - 4 = [x^2 - y^2 - 4y - 4] + i[2xy + 4x].$$

We have

$$\begin{aligned}u(x, y) &= \operatorname{Re} f(z) = x^2 - y^2 - 4y - 4, \\ v(x, y) &= \operatorname{Im} f(z) = 2xy + 4x.\end{aligned}$$

The partial derivatives are

$$u_x = 2x, \quad v_x = 2x, \quad u_y = -2y - 4, \quad v_y = 2y + 4.$$

We see that

$$u_x = v_y, \quad u_y = -v_x.$$

Problem 9 At which points $z \in \mathbb{C}$ is the function $f(z) = z|z|$ differentiable? Where is this function holomorphic?

Solution.

We know that the function $z \mapsto |z|$ is not differentiable at any point $z \in \mathbb{C}$. If $z \neq 0$, then $|z| = \frac{f(z)}{z}$, and the function $z \mapsto \frac{1}{z}$ is differentiable. It follows that $f(z)$ is not differentiable for $z \neq 0$ (otherwise $|z|$ were differentiable as a product of two differentiable functions).

Alternatively, this can be shown using the Cauchy-Riemann equations. Indeed, for $z = x + iy$ we have

$$f(z) = (x + iy)\sqrt{x^2 + y^2},$$

so that

$$u(x, y) = x\sqrt{x^2 + y^2}, \quad v(x, y) = y\sqrt{x^2 + y^2}.$$

The partial derivatives are

$$u_x = \frac{2x^2 + y^2}{\sqrt{x^2 + y^2}}, \quad v_y = \frac{x^2 + 2y^2}{\sqrt{x^2 + y^2}}, \quad u_y = \frac{xy}{\sqrt{x^2 + y^2}}, \quad v_x = \frac{xy}{\sqrt{x^2 + y^2}}.$$

The Cauchy-Riemann equations $u_x = v_y$, $u_y = -v_x$ imply that

$$2x^2 + y^2 = x^2 + 2y^2, \quad xy = -xy.$$

The only solution of this system of equations is $x = 0$, $y = 0$. Consequently, f is not differentiable at any $z \neq 0$.

Now let us consider the point $z = 0$. We consider the limit

$$\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h|h| - 0}{h} = \lim_{h \rightarrow 0} |h| = 0.$$

The limit exists; this means that f is differentiable at $z = 0$ with $f'(0) = 0$.

The function $f(z) = z|z|$ is nowhere holomorphic since there is no open set where f is differentiable.

Problem 10 Where is the function $f(z) = \frac{1}{z^2-1}$ holomorphic?

Solution.

The function $z \mapsto z^2 - 1$ is differentiable everywhere since it is a polynomial. It follows that the function $f(z) = \frac{1}{z^2-1}$ is differentiable on its entire domain $\mathbb{C} \setminus \{-1, 1\}$. The latter set is open. It follows that f is holomorphic on $\mathbb{C} \setminus \{-1, 1\}$.